

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 6: Measuring Growth: Regions of Growth and Growth Curves of Maize H614 and Bean Rosecoco

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	D	B	B	B	D	A	D

USE AFTER: OBSERVE PHASE — Part A: Regions of growth (ink marks on maize and bean roots)

Q1. Fatuma's group marks a bean root with ink every 2 mm from the tip for 10 mm. After two days, the marks nearest the tip have spread furthest apart, while the marks further back are still close together. What does this show?

- A. The root grows in length mainly from the base near the seed
- B. The root grows in length equally along its whole length
- C. The root grows in length mainly in a region just behind the tip**
- D. The root grows only in thickness and not in length

Correct answer: C. Marks near the tip moved apart most, so cells are elongating just behind the root tip. Marks further back stayed close together, so those regions are not lengthening much.

USE AFTER: EXPLAIN PHASE — Explain: CER linking marks spreading near the tip to the apical meristem

Q2. A student says: 'Marks spreading near the root tip support the idea that new cells are made at the apical meristem and then elongate.' Which order of regions, starting from the very tip of a root, is correct?

- A. Region of elongation, region of cell division, region of differentiation
- B. Region of differentiation, region of elongation, region of cell division
- C. Region of cell division, region of differentiation, region of elongation
- D. Region of cell division, region of elongation, region of differentiation**

Correct answer: D. Cells divide at the apical meristem at the tip. They then elongate behind it and later differentiate into specialised cells further back.

USE AFTER: OBSERVE PHASE — Part B: calculating the increase in height between each pair of measurements

Q3. The mean shoot height of maize H614 was 62 cm at Day 28 and 98 cm at Day 35. What was the increase in height between Day 28 and Day 35?

- A. 160 cm
- B. 36 cm**
- C. 27 cm
- D. 32 cm

Correct answer: B. Increase = $98 - 62 = 36$ cm. The answer 32 cm is the increase from Day 21 to Day 28, and 27 cm is the increase from Day 35 to Day 42.

USE AFTER: OBSERVE PHASE — Part B: 'Between which days did bean increase most?'

Q4. The mean height of bean Rosecoco was 4, 9, 18, 30, 40 and 44 cm at Days 7, 14, 21, 28, 35 and 42. Between which two measurements was the increase in height greatest?

- A. Day 14 and Day 21
- B. Day 21 and Day 28**
- C. Day 35 and Day 42
- D. Day 7 and Day 14

Correct answer: B. The increases are 5, 9, 12, 10 and 4 cm. The greatest is 12 cm ($30 - 18$) between Day 21 and Day 28, which is within the steepest part of the curve.

USE AFTER: OBSERVE PHASE — Part B: drawing smooth curves and shading the lag, log and stationary phases

Q5. Maize H614 increased by only 9 cm between Day 7 and Day 14 (3 cm to 12 cm), but by 36 cm between Day 28 and Day 35. Which phase of the growth curve does the first period show?

- A. Stationary phase, because growth has stopped
- B. Lag phase, because growth is slow at the start**
- C. Death phase, because the plant is weak
- D. Log phase, because growth is fastest

Correct answer: B. The lag phase is the slow start of growth, when increases are small. The log phase, with the fastest growth, comes later.

USE AFTER: EXPLAIN PHASE — Explain: puzzle 'Why does the bean slow down by Day 42 while maize is still taller?'

Q6. Bean Rosecoco was 40 cm tall at Day 35 and 44 cm at Day 42, so its curve is flattening. What does the flattening show?

- A. The bean plant has died
- B. The bean plant is in the log phase, where growth is fastest
- C. The bean plant is starting its lag phase
- D. The bean plant is in the stationary phase, where growth slows or stops**

Correct answer: D. A flattening curve means growth has slowed, which is the stationary phase. It does not mean the plant is dead, which is a common misconception.

USE AFTER: EXPLAIN PHASE — Explain: probe 'Does a taller plant at Day 42 mean it grew faster all the time?'

Q7. At Day 7, bean Rosecoco was 4 cm tall and maize H614 was 3 cm. At Day 42, maize was 125 cm and bean was 44 cm. A student says: 'Maize must have grown faster than bean all the time.' Which response is correct?

- A. Wrong, because bean was taller than maize at Day 7, so maize did not grow faster all the time**
- B. Correct, because maize is taller at Day 42
- C. Wrong, because bean grew faster than maize between Day 28 and Day 42
- D. Correct, because maize roots are longer than bean roots

Correct answer: A. Bean was taller at Day 7 (4 cm against 3 cm). Maize overtook it from Day 14 onward, so being taller at the end does not mean faster growth at every stage.

USE AFTER: MODEL BUILDING PHASE — Model: adding a small sketch of the sigmoid growth curve with lag, log and stationary phases

Q8. A learner draws a graph of shoot height (y-axis) against time in days (x-axis) for maize H614 over 42 days. Which description matches the overall shape of a normal growth curve?

- A. A curve that rises steeply at first and then falls back to zero
- B. A horizontal line
- C. A straight line sloping upwards
- D. An S-shaped (sigmoid) curve with lag, log and stationary phases**

Correct answer: D. A growth curve is sigmoid. It starts slowly (lag), rises steeply (log) and then levels off (stationary). It does not rise in a straight line or fall back down.